

Лабораторная работа № 6

Основы интерфейса взаимодействия пользователя с системой
Unix на уровне командной строки

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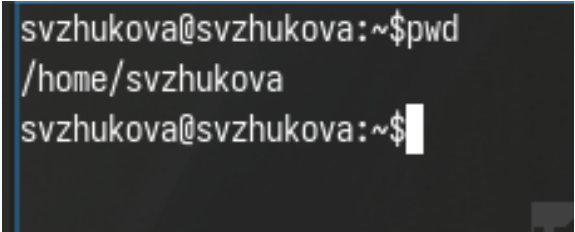
Вводная часть

Цели и задачи

Приобретение практических навыков взаимодействия пользователя с системой по- средством командной строки.

Выполнение лабораторной работы

Определим полное имя нашего домашнего каталога.

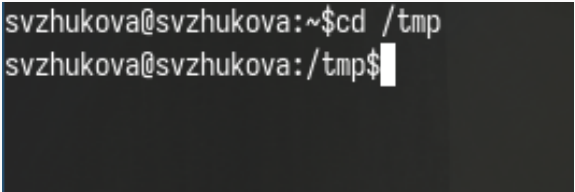


```
svzhukova@svzhukova:~$pwd  
/home/svzhukova  
svzhukova@svzhukova:~$
```

A terminal window with a dark background and light blue text. The prompt is 'svzhukova@svzhukova:~\$'. The user enters 'pwd', and the output is '/home/svzhukova'. The prompt then changes to 'svzhukova@svzhukova:~\$' with a white cursor block.

Figure 1: Имя нашего домашнего каталога

Перейдем в каталог /tmp.



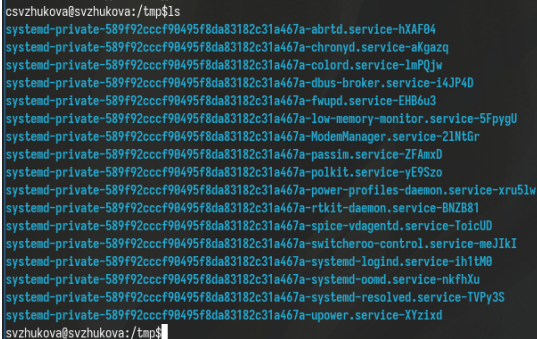
```
svzhukova@svzhukova:~$cd /tmp  
svzhukova@svzhukova:/tmp$
```

A terminal window with a dark background and light blue text. The prompt is 'svzhukova@svzhukova:~\$'. The user enters 'cd /tmp', and the output is 'svzhukova@svzhukova:/tmp\$'. The prompt then changes to 'svzhukova@svzhukova:/tmp\$' with a white cursor block.

Выведите на экран содержимое каталога `/tmp`. Для этого используйте команду `ls`

с различными опциями.

Команда `ls` используется для просмотра содержимого каталога



```
csvzhukova@svzhukova: /tmp$ ls
systemd-private-589f92cccf90495f8da83182c31a467a-abrt.d.service-hXAF04
systemd-private-589f92cccf90495f8da83182c31a467a-chronyd.service-aKgazq
systemd-private-589f92cccf90495f8da83182c31a467a-colord.service-lmPQjw
systemd-private-589f92cccf90495f8da83182c31a467a-dbus-broker.service-i4JP4D
systemd-private-589f92cccf90495f8da83182c31a467a-fwupd.service-EHB6u3
systemd-private-589f92cccf90495f8da83182c31a467a-low-memory-monitor.service-5FpygU
systemd-private-589f92cccf90495f8da83182c31a467a-ModemManager.service-2lNtGr
systemd-private-589f92cccf90495f8da83182c31a467a-passim.service-ZFamxD
systemd-private-589f92cccf90495f8da83182c31a467a-polkit.service-yE9Szo
systemd-private-589f92cccf90495f8da83182c31a467a-power-profiles-daemon.service-xru5lw
systemd-private-589f92cccf90495f8da83182c31a467a-rtkit-daemon.service-BNZB81
systemd-private-589f92cccf90495f8da83182c31a467a-spice-vdagentd.service-ToicUD
systemd-private-589f92cccf90495f8da83182c31a467a-switcheroo-control.service-meJikI
systemd-private-589f92cccf90495f8da83182c31a467a-systemd-logind.service-ih1tM0
systemd-private-589f92cccf90495f8da83182c31a467a-systemd-oomd.service-nkFhXu
systemd-private-589f92cccf90495f8da83182c31a467a-systemd-resolved.service-TVPy3S
systemd-private-589f92cccf90495f8da83182c31a467a-upower.service-XYzixd
svzhukova@svzhukova: /tmp$
```

Figure 3: `ls`

Для того, чтобы отобразить имена скрытых файлов, необходимо использовать команду `ls` с опцией `a`

```
svzhukova@svzhukova: /tmp$ ls -la
ls
ls -lRt -uall
ls -lRt -uall
system-private-589f92ccc9b495f8da3182c31a467a-avrdt.service-hIAF84
system-private-589f92ccc9b495f8da3182c31a467a-chromyd.service-akgqz
system-private-589f92ccc9b495f8da3182c31a467a-colord.service-lmPQjw
system-private-589f92ccc9b495f8da3182c31a467a-dbus-broker.service-14JP4D
system-private-589f92ccc9b495f8da3182c31a467a-fwupd.service-EH8u3
system-private-589f92ccc9b495f8da3182c31a467a-low-memory-monitor.service-5fpgU
system-private-589f92ccc9b495f8da3182c31a467a-ModemManager.service-21Nt5r
system-private-589f92ccc9b495f8da3182c31a467a-passin.service-2FAmD
system-private-589f92ccc9b495f8da3182c31a467a-polkit.service-yE9Szo
system-private-589f92ccc9b495f8da3182c31a467a-power-profiles-daemon.service-xru5lw
svzhukova@svzhukova: /tmp$
```

Figure 4: `ls -la`

Чтобы вывести на экран подробную информацию о файлах и каталогах, необходимо использовать опцию `l`

```
svzhukova@svzhukova: /tmp$ ls -l
ls
ls -lRt -uall
ls -lRt -uall
drwx----- 3 root root 68 map 19 14:55 system-private-589f92ccc9b495f8da3182c31a467a-avrdt.service-hIAF84
drwx----- 3 root root 68 map 19 14:55 system-private-589f92ccc9b495f8da3182c31a467a-chromyd.service-akgqz
drwx----- 3 root root 68 map 19 14:56 system-private-589f92ccc9b495f8da3182c31a467a-colord.service-lmPQjw
drwx----- 3 root root 68 map 19 14:55 system-private-589f92ccc9b495f8da3182c31a467a-dbus-broker.service-14JP4D
drwx----- 3 root root 68 map 19 14:56 system-private-589f92ccc9b495f8da3182c31a467a-fwupd.service-EH8u3
drwx----- 3 root root 68 map 19 14:55 system-private-589f92ccc9b495f8da3182c31a467a-low-memory-monitor.service-5fpgU
drwx----- 3 root root 68 map 19 14:55 system-private-589f92ccc9b495f8da3182c31a467a-ModemManager.service-21Nt5r
drwx----- 3 root root 68 map 19 14:56 system-private-589f92ccc9b495f8da3182c31a467a-passin.service-2FAmD
drwx----- 3 root root 68 map 19 14:55 system-private-589f92ccc9b495f8da3182c31a467a-polkit.service-yE9Szo
drwx----- 3 root root 68 map 19 14:56 system-private-589f92ccc9b495f8da3182c31a467a-power-profiles-daemon.service-xru5lw
drwx----- 3 root root 68 map 19 14:55 system-private-589f92ccc9b495f8da3182c31a467a-spice-vdagentd.service-toicUD
drwx----- 3 root root 68 map 19 14:56 system-private-589f92ccc9b495f8da3182c31a467a-switcheroo-control.service-maJk1
drwx----- 3 root root 68 map 19 14:55 system-private-589f92ccc9b495f8da3182c31a467a-system-logind.service-ihUW
drwx----- 3 root root 68 map 19 14:55 system-private-589f92ccc9b495f8da3182c31a467a-system-oomd.service-nkfhU
```

Можно также получить информацию о типах файлов (каталог, исполняемый файл, ссылка), для чего используется опция F

```
svzhukova@svzhukova:/tmp$ ls -F
system-private-589f2ccc9b495f8da83182c31a467a-abrttd.service-hAF8u/
system-private-589f2ccc9b495f8da83182c31a467a-chromyd.service-akgzu/
system-private-589f2ccc9b495f8da83182c31a467a-colord.service-lmQJw/
system-private-589f2ccc9b495f8da83182c31a467a-dbus-broker.service-lAP40/
system-private-589f2ccc9b495f8da83182c31a467a-fuupd.service-ENB6u/
system-private-589f2ccc9b495f8da83182c31a467a-low-memory-monitor.service-5fpygl/
system-private-589f2ccc9b495f8da83182c31a467a-NodeManager.service-2lM6r/
system-private-589f2ccc9b495f8da83182c31a467a-passia.service-27Am0/
system-private-589f2ccc9b495f8da83182c31a467a-power-profiles-daemon.service-xru5w/
system-private-589f2ccc9b495f8da83182c31a467a-rtkit-daemon.service-BN28t/
system-private-589f2ccc9b495f8da83182c31a467a-qt5c-vgemtd.service-folc0/
system-private-589f2ccc9b495f8da83182c31a467a-switcheroo-control.service-mkHxU/
system-private-589f2ccc9b495f8da83182c31a467a-system-logind.service-jhDB6/
system-private-589f2ccc9b495f8da83182c31a467a-system-ondemand.service-mfHxU/
system-private-589f2ccc9b495f8da83182c31a467a-system-resolved.service-TVP35/
system-private-589f2ccc9b495f8da83182c31a467a-upower.service-X7rzd/
```

Figure 6: ls -F

Оптимизированная команда ls с различными опциями

```
svzhukova@svzhukova:/tmp$ ls -alF
итого 28
drwxrwxrwt. 23 root root 568 map 19 15:08 ./
dr-xr-xr-x. 1 root root 158 map 13 18:19 ../
drwxrwxrwt. 2 root root 48 map 19 14:55 font-units/
drwxrwxrwt. 2 root root 88 map 19 14:56 lxi-unix/
drwx----- 3 root root 68 map 19 14:55 system-private-589f2ccc9b495f8da83182c31a467a-abrttd.service-hAF8u/
drwx----- 3 root root 68 map 19 14:55 system-private-589f2ccc9b495f8da83182c31a467a-chromyd.service-akgzu/
drwx----- 3 root root 68 map 19 14:56 system-private-589f2ccc9b495f8da83182c31a467a-colord.service-lmQJw/
drwx----- 3 root root 68 map 19 14:55 system-private-589f2ccc9b495f8da83182c31a467a-dbus-broker.service-lAP40/
drwx----- 3 root root 68 map 19 14:56 system-private-589f2ccc9b495f8da83182c31a467a-fuupd.service-ENB6u/
drwx----- 3 root root 68 map 19 14:55 system-private-589f2ccc9b495f8da83182c31a467a-low-memory-monitor.service-5fpygl/
drwx----- 3 root root 68 map 19 14:55 system-private-589f2ccc9b495f8da83182c31a467a-NodeManager.service-2lM6r/
drwx----- 3 root root 68 map 19 14:56 system-private-589f2ccc9b495f8da83182c31a467a-passia.service-27Am0/
drwx----- 3 root root 68 map 19 14:55 system-private-589f2ccc9b495f8da83182c31a467a-polkitt.service-yE9z2o/
drwx----- 3 root root 68 map 19 14:56 system-private-589f2ccc9b495f8da83182c31a467a-power-profiles-daemon.service-xru5w/
drwx----- 3 root root 68 map 19 14:55 system-private-589f2ccc9b495f8da83182c31a467a-rtkit-daemon.service-BN28t/
drwx----- 3 root root 68 map 19 14:56 system-private-589f2ccc9b495f8da83182c31a467a-qt5c-vgemtd.service-folc0/
drwx----- 3 root root 68 map 19 14:55 system-private-589f2ccc9b495f8da83182c31a467a-switcheroo-control.service-mkHxU/
drwx----- 3 root root 68 map 19 14:55 system-private-589f2ccc9b495f8da83182c31a467a-system-logind.service-jhDB6/
drwx----- 3 root root 68 map 19 14:55 system-private-589f2ccc9b495f8da83182c31a467a-system-ondemand.service-mfHxU/
drwx----- 3 root root 68 map 19 14:55 system-private-589f2ccc9b495f8da83182c31a467a-system-resolved.service-TVP35/
drwx----- 3 root root 68 map 19 14:55 system-private-589f2ccc9b495f8da83182c31a467a-upower.service-X7rzd/
-r--r--r-- 1 svzhukova svzhukova 11 map 19 14:56 .X0-lock
-r--r--r-- 1 gdm gdm 11 map 19 14:56 .X1824-lock
-r--r--r-- 1 gdm gdm 11 map 19 14:56 .X1825-lock
drwxrwxrwt. 2 root root 148 map 19 14:56 x11-unix/
-r--r--r-- 1 svzhukova svzhukova 11 map 19 14:56 .X1-lock
-r--r--r-- 1 svzhukova svzhukova 11 map 19 14:56 .X2-lock
drwxrwxrwt. 2 root root 48 map 19 14:55 xlx-unix/
svzhukova@svzhukova:/tmp$
```

Определим, есть ли в каталоге `/var/spool` подкаталог с именем `cron`?

```
svzhukova@svzhukova:~$ ls /var/spool/
abrt  abrt-upload  cups  lpd  mail  plymouth
```

Figure 8: Каталога `cron` нет

Перейдем в наш домашний каталог и выведем на экран его содержимое.

```
svzhukova@svzhukova:~$ ls -l
total 88
-rw-r--r--. 1 svzhukova svzhukova 47187 мар 15 01:26 32.png
-rw-r--r--. 1 svzhukova svzhukova 65 мар 2 15:20 95-system-keyboard-config.conf
drwxr-xr-x. 1 svzhukova svzhukova 14 мар 13 23:14 bin
drwxr-xr-x. 1 svzhukova svzhukova 0 мар 13 23:25 Documents
drwxr-xr-x. 1 svzhukova svzhukova 814 мар 19 14:40 Downloads
-rw-r--r--. 1 svzhukova svzhukova 67 фев 23 20:12 exec
drwxr-xr-x. 1 svzhukova svzhukova 74 мар 3 17:08 git-extended
drwxr-xr-x. 1 svzhukova svzhukova 24 мар 13 23:16 git@github.com:svzhukova
-rw-r--r--. 1 svzhukova svzhukova 18657 мар 13 23:04 LICENSE
drwxr-xr-x. 1 svzhukova svzhukova 250 мар 1 19:53 os-intro
drwxr-xr-x. 1 svzhukova svzhukova 50 мар 13 13:50 pass
drwxr-xr-x. 1 svzhukova svzhukova 50 мар 17 19:18 Pictures
-rw-r--r--. 1 svzhukova svzhukova 63 мар 2 14:29 tt
-rw-r--r--. 1 svzhukova svzhukova 67 мар 2 14:44 ttt
-rw-r--r--. 1 svzhukova svzhukova 63 мар 2 14:42 tttt
drwxr-xr-x. 1 svzhukova svzhukova 50 мар 2 17:17 work
drwxr-xr-x. 1 svzhukova svzhukova 0 фев 22 20:19 Видео
drwxr-xr-x. 1 svzhukova svzhukova 0 фев 22 20:19 Документы
drwxr-xr-x. 1 svzhukova svzhukova 726 мар 7 19:33 Загрузки
```

Выполним следующие действия

В домашнем каталоге создадим новый каталог с именем newdir.

```
svzhukova@svzhukova:~$mkdir newdir
```

Figure 10: Каталог с именем newdir

В каталоге ~/newdir создадим новый каталог с именем morefun.

```
svzhukova@svzhukova:~$mkdir newdir/morefun
svzhukova@svzhukova:~$cd newdir
svzhukova@svzhukova:~/newdir$ls
morefun
svzhukova@svzhukova:~/newdir$
```

Figure 11: Каталог с именем morefun

В домашнем каталоге создадим одной командой три новых каталога с именами

letters, memos, misk.

```
svzhukova@svzhukova:~$mkdir letters memos misk
```

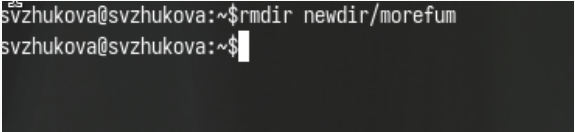
Figure 12: letters, memos, misk

```
svzhukova@svzhukova:~$rmrmdir letter memos misk
rmrmdir: не удалось удалить 'letter': Нет такого файла или каталога
svzhukova@svzhukova:~$rmrmdir letters memos misk
rmrmdir: не удалось удалить 'memos': Нет такого файла или каталога
rmrmdir: не удалось удалить 'misk': Нет такого файла или каталога
svzhukova@svzhukova:~$
```

Figure 13: Удалим letters, memos, misk

Попробуем удалить ранее созданный каталог ~/newdir командой rm.

Удалим каталог ~/newdir/morefun из домашнего каталога.

A terminal window with a dark background. The prompt is 'svzhukova@svzhukova:~\$'. The command 'rm -r newdir/morefun' is entered. The prompt returns as 'svzhukova@svzhukova:~\$' with a cursor at the end.

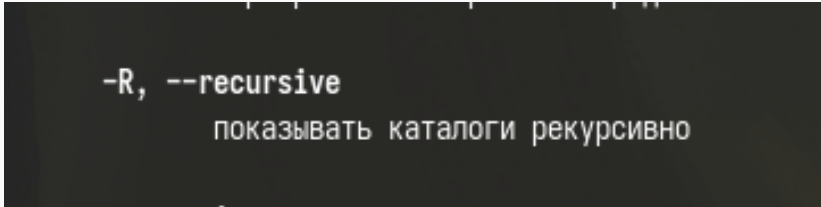
```
svzhukova@svzhukova:~$rm -r newdir/morefun
svzhukova@svzhukova:~$
```

Figure 14: Удалим каталог ~/newdir/morefun из домашнего каталога

С помощью команды `man` определим, какую опцию команды `ls` нужно использо-

вать для просмотра содержимое не только указанного каталога, но и подкаталогов, входящих в него.

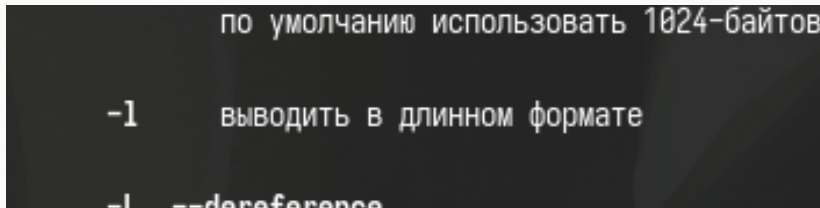
`-R`

A screenshot of a terminal window showing the man page for the 'ls' command. The text '-R, --recursive' is highlighted in a light blue box, and the Russian translation 'показывать каталоги рекурсивно' is shown below it in a light green box. The background is dark gray with white text.

`-R, --recursive`
показывать каталоги рекурсивно

Figure 15: Поиск нужной опции

С помощью команды `man ls` определяем набор опций команды, позволяющий отсортировать по времени последнего изменения выводимый список содержимого каталога. Таким набором опций являются: `-c`

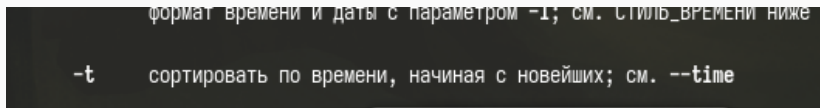


по умолчанию использовать 1024-байтов

-l выводить в длинном формате

-l `--dereference`

Figure 16: `-c`



формат времени и даты с параметром `-l`; см. СТИЛЬ_ВРЕМЕНИ ниже

-t сортировать по времени, начиная с новейших; см. `--time`

Проверим работу команды

```
svzhukova@svzhukova:~$ls -tl
итого 88
drwxr-xr-x. 1 svzhukova svzhukova    0 map 19 15:24 newdir
drwxr-xr-x. 1 svzhukova svzhukova   814 map 19 14:40 Downloads
drwxr-xr-x. 1 svzhukova svzhukova    50 map 17 19:18 Pictures
-rw-r--r--. 1 svzhukova svzhukova 47107 map 15 01:26 32.png
drwxr-xr-x. 1 svzhukova svzhukova    0 map 13 23:25 Documents
drwxr-xr-x. 1 svzhukova svzhukova    24 map 13 23:16 git@github.com:svzhukova
drwxr-xr-x. 1 svzhukova svzhukova    14 map 13 23:14 bin
-rw-r--r--. 1 svzhukova svzhukova 18657 map 13 23:04 LICENSE
drwxr-xr-x. 1 svzhukova svzhukova    50 map 13 13:50 pass
drwxr-xr-x. 1 svzhukova svzhukova    726 map 7 19:33 Загрузки
drwxr-xr-x. 1 svzhukova svzhukova    0 map 4 22:20 Музыка
drwxr-xr-x. 1 svzhukova svzhukova    74 map 3 17:08 git-extended
drwxr-xr-x. 1 svzhukova svzhukova    50 map 2 17:17 work
-rw-r--r--. 1 svzhukova svzhukova    65 map 2 15:20 95-system-keyboard-config.conf
-rw-r--r--. 1 svzhukova svzhukova    67 map 2 14:44 ttt
-rw-r--r--. 1 svzhukova svzhukova    63 map 2 14:42 tttt
-rw-r--r--. 1 svzhukova svzhukova    63 map 2 14:29 tt
drwxr-xr-x. 1 svzhukova svzhukova   250 map 1 19:53 os-intro
drwxr-xr-x. 1 svzhukova svzhukova    50 фев 25 22:45 Изображения
-rw-r--r--. 1 svzhukova svzhukova    67 фев 23 20:12 ехес
drwxr-xr-x. 1 svzhukova svzhukova    0 фев 22 20:19 Видео
drwxr-xr-x. 1 svzhukova svzhukova    0 фев 22 20:19 Документы
drwxr-xr-x. 1 svzhukova svzhukova    0 фев 22 20:19 Общедоступные
drwxr-xr-x. 1 svzhukova svzhukova    0 фев 22 20:19 'Рабочий стол'
drwxr-xr-x. 1 svzhukova svzhukova    0 фев 22 20:19 Шаблоны
svzhukova@svzhukova:~$
```

Используем команду `man` для просмотра описания следующих команд:
`cd`, `pwd`, `mkdir`,

`rmdir`, `rm`.

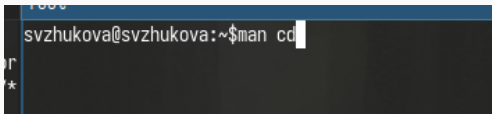


Figure 19: `man cd`

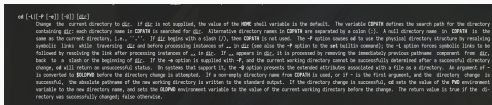


Figure 20: `man cd`

```
svzhukova@svzhukova:~$man pwd
```

Figure 21: man pwd

```
PWD(1)                                User Commands                                PWD(1)

NAME
    pwd - print name of current/working directory

SYNOPSIS
    pwd [OPTION]...

DESCRIPTION
    Print the full filename of the current working directory.

    -L, --logical
        use PWD from environment, even if it contains symlinks

    -P, --physical
        avoid all symlinks

    --help display this help and exit

    --version
        output version information and exit

    If no option is specified, -P is assumed.

NOTE: your shell may have its own version of pwd, which usually supersedes the version described here. Please refer to your
shell's documentation for details about the options it supports.

AUTHOR
    Written by Jim Meyering.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

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SEE ALSO
    getcwd(3)

    Full documentation <https://www.gnu.org/software/coreutils/pwd>
```

```
svzhukova@svzhukova:~$man mkdir
```

Figure 23: man mkdir

```
mkdir(1)                                User Contributed                  mkdir(1)

NAME
  mkdir - make directories

SYNOPSIS
  mkdir [OPTION]... DIRECTORY...

DESCRIPTION
  Create the DIRECTORY(ies), if they do not already exist.

  Mandatory arguments to long options are mandatory for short options too.

  -h, --help
      get this manual page in stdin, not stderr - useful

  -p, --parents
      no error if existing, make parent directories as needed, with their file modes unaffected by any -m option.

  -v, --verbose
      print a message for each created directory

  -Z
      set file's security context of each created directory to the default type

  --context=CTX
      like -Z, or if CTX is specified then set the SELinux or SMACK security context to CTX

  --help
      display this help and exit

  --version
      output version information and exit

AUTHOR
  Written by David Reedsen.

REPORTING BUGS
  Report bugs to https://www.gnu.org/software/coreutils/.
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SEE ALSO
  mkdir(2)

Full documentation: https://www.gnu.org/software/coreutils/mkdir.html
or available locally via: info:linuxinfo/mkdir/mkdir.html

GNU coreutils 9.4                               November 2023                               mkdir(1)
Bash: page mkdir(1) (line 174) (GNU) (press h for help or q to exit)
```

Figure 24: man mkdir

```
svzhukova@svzhukova:~$man rmdir
```

Figure 25: man rmdir

```
RMDIR(1)                                User Commands                                RMDIR(1)

NAME
    rmdir - remove empty directories

SYNOPSIS
    rmdir [OPTION]... DIRECTORY...

DESCRIPTION
    Remove the DIRECTORY(ies), if they are empty.

    --ignore-fail-on-non-empty
        ignore each failure to remove a non-empty directory

    -p, --parents
        remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b a'

    -v, --verbose
        output a diagnostic for every directory processed

    --help
        display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David McKenzie.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
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SEE ALSO
    rmdir(2)

    Full documentation <https://www.gnu.org/software/coreutils/rmdir>
    or available locally via: info '(coreutils) rmdir invocation'
```

```
svzhukova@svzhukova:~$man rm
```

Figure 27: man rm

```
NAME
rm - remove files or directories

SYNOPSIS
rm [OPTION]... [FILE]...

DESCRIPTION
This manual page documents the GNU version of rm.  rm removes each specified file.  By default, it does not remove directories.

If the -i or --interactive option is given, and there are more than three files or the -r, -d, or --recursive are given, then rm prompts the user for whether to proceed with the entire operation.  If the response is not affirmative, the entire command is aborted.

Otherwise, if a file is writable, standard input is a terminal, and the -i or --interactive option is not given, or the -i or --interactive option is given, rm prompts the user for whether to remove the file.  If the response is not affirmative, the file is skipped.

OPTIONS
Remove (unlink) the file(s).

-f, --force
    ignore nonexistent files and arguments, never prompt

-i
    prompt before every removal

-I
    prompt once before removing more than three files, or when removing recursively; less intrusive than -i, while still giving protection against most mistakes

--interactive[=MODE]
    prompt according to MODE: never, once (-I), or always (-i); without MODE, prompt always

--one-file-system
    when removing a hierarchy recursively, skip any directory that is on a file system different from that of the corresponding command line argument

--no-preserve-root
    do not treat '/' specially

--preserve-root[=ALL]
    do not remove '/' (default); with 'all', reject any command line argument on a separate device from its parent

-r, -R, --recursive
    remove directories and their contents recursively

-d, --dir
    remove empty directories

-v, --verbose
    explain what is being done

--help
    display this help and exit

--version
    output version information and exit

By default, rm does not remove directories.  Use the --recursive (-r or -R) option to remove each listed directory, too, along with all of its contents.

To remove a file whose name starts with a '-', for example '-foo', use one of these commands:

rm -- -foo

rm ./-foo
```

Используя информацию,

полученную при помощи команды history, выполним модификацию и исполнение нескольких команд из буфера команд.

```
root@kali:~# history
1  ls
2  sudo def install smp saylock
3  sudo def install smp saylock vi-clipboard smp pris slarp
4  ls
5  sudo pacman -S networkmanager
6  sudo def install smp
7  sudo def install smpmg
8  sudo def install smp saymg
9  smp output " enable
10 sudo su
11 nano
12 nano ~/.bashrc
13 source ~/.bashrc
14 echo $PS1$PROMPT$
15 ls
16 sudo def install smp
17 sudo def install smp-all-server sayland
18 mkdir -p ~/.config/smp
19 touch ~/.config/smp/config
20 nano ~/.config/smp/config
21 ls
22 sudo def install smp-all-server sayland sayland-protocols
23 ls
24 sudo def install smp-config fedora
25 sudo def gpg install smp-desktop-environment
26 sudo def rmconf smp-config-system smp-config-fedora
27 sudo def install smp-config-fedora
28 sudo def upgrade --refresh
29 ls
30 ls -l
31 gkill smp
32 ls
33 sudo systemctl restart smp
34 ls
35 rpm -q smp
36 starts
37 sudo smp
38 sudo smp
39 sudo def install alacritty
40 ls
41 rm
42 ls
43 sudo def install smp-dm smp-vllm-drivers
44 sudo def install smp-dm-drivers smp-vllm-drivers
45 sudo def install sayland sayland-tilde
46 sudo def install sayland-devel
47 sudo def install libsayland
48 sudo def makecmw
49 sudo def install sayland-protocols
50 sudo def install sayland-devel
51 sudo def install sayland-devel
52 def register
53 sudo def gpginstall "Development Tools"
54 sudo def gpginstall "C Development Tools and Libraries"
55 sudo def update
56 ls
57 sudo def provide sayland
58 rpm -qa | grep sayland
59 sudo def install sayland
60 sudo def install sayland sayland-protocols sayland-cursor-themes
61 ls
62 sudo def install sayland-protocols-devel
63 sudo def install sayland-devel
64 sudo rmconf
65 ls
66 nano ~/.config/smp/config
67 ls
68 [B]rowse ~/.config/smp/config
69 nano ~/.config/smp/config
70 ls
71 sudo i
72 sudo -i
```

```
svzhukova@svzhukova:~$!992:s/morefum/  
rmdir newdir/  
svzhukova@svzhukova:~$
```

Figure 30: Модификация и исполнение команд

Мы приобрели практически навыки взаимодействия пользователя с системой по- средством командной строки.